

KENYA MAIZE PRODUCTION & CLIMATE DATA ANALYSIS

A Quantitative Analysis of Maize Production Trends and Climate Variable Relationships (2000–2024)

Author:	Independent Personal Project	Data Period:	2000 – 2024 (25 Years)
Data Sources:	FAOSTAT & World Bank Climate Change Knowledge Portal (CCKP)	Status:	Final Analysis Report

Executive Summary

This personal project report presents a quantitative evaluation of maize production trends in Kenya between 2000 and 2024, examining relationships between agricultural outputs and key climate parameters. By integrating national-level production, yield, and harvested-area data from **FAOSTAT** with climate data (temperature, precipitation, and growing-season indicators) from the **World Bank Climate Change Knowledge Portal (CCKP)**, this study characterizes temporal dynamics and structural drivers of production growth.

Key Findings: Total maize production grew by **86.25%** over the 25-year period (from 2.16M tonnes in 2000 to 4.285M tonnes in 2023). A linear decomposition demonstrates that **70.68%** of this growth was driven by *harvested area expansion*, whereas *direct yield improvements* contributed **18.21%**. Correlation analysis identifies a moderate positive association between precipitation and yield ($r = 0.61$), underscoring the rainfed sector's ongoing climate exposure.

1. Introduction & Project Scope

1.1 Background

Maize (*Zea mays L.*) is the foundational staple crop in Kenya's agricultural economy and household food security system. Serving as the primary calorie source for millions of households, maize dynamics heavily influence national food prices, rural incomes, and overall economic stability. Agricultural performance remains tightly coupled with physical climate inputs—principally precipitation amounts and temperature regimes during key crop development phases.

1.2 Project Objectives

This independent data analytics project was established to achieve six specific quantitative objectives across the 2000–2024 period:

- Analyze temporal trends in total maize production, yield per hectare, and harvested land area.
- Characterize overall historical variations in temperature, precipitation, and planting dates.
- Quantify linear associations between climate variables and maize crop yields (**kg/ha**).
- Evaluate associations between climate inputs and gross national maize production.
- Identify historical outlier years characterized by peak productivity or major yield shortfalls.

- Decompose gross production changes between 2000 and 2024 into distinct area and yield components.

1.3 Methodological Scope & Limitations

The dataset comprises 25 annual observations (2000–2024) across 8 integrated variables. Data quality checks confirmed complete records with zero missing values or duplicate rows. Primary climate parameters were sourced from the World Bank Climate Change Knowledge Portal (CCKP). Analytical methods include descriptive metrics, Pearson correlation, IQR-based outlier checks, mathematical production decomposition, and interactive visualization via Power BI.

Analytical Limitations: Analysis is restricted to national aggregate figures without regional/county breakdown; correlation metrics denote association rather than proven causality; aggregate annual climate metrics may mask localized intra-seasonal dry spells; and inferential predictive modeling is beyond the current exploratory scope.

2. Quantitative Findings & Analysis

2.1 Baseline Descriptive Statistics

Table 1 summarizes the baseline statistical parameters across all physical and climatic metrics analyzed over the 25-year timeframe.

Variable	Mean	Std Dev	Min	Q1	Median	Q3
Yield (kg/ha)	1,630.15	154.80	1,294.30	1,523.80	1,660.20	1,736.60
Production (tonnes)	3,227,646	576,761	2,160,000	2,790,000	3,304,430	3,592,688
Harvested Area (ha)	1,980,250	300,195	1,351,327	1,700,000	2,098,240	2,159,322
Temperature (°C)	25.51	0.28	25.03	25.29	25.52	25.75
Precipitation (mm)	678.21	136.30	423.31	597.52	687.12	741.30

Annual maize yield averaged **1,630.15~kg/ha**, fluctuating between a minimum of **1,294.3~kg/ha** and a peak of over **1,763~kg/ha**. Gross annual production averaged 3.23 million tonnes. Mean annual temperatures exhibited high stability, whereas annual precipitation exhibited substantial variability ranging from 423.31 mm to 948 mm.

2.2 Temporal Trend & Production Growth

Between 2000 and 2024, national maize output expanded substantially. Table 2 details the comparative shifts across key production parameters.

Metric	Year 2000 Baseline	Year 2024 Level	Absolute Shift	Percentage Change
Production (tonnes)	2,160,000	4,023,000	+1,863,000	+86.25%
Yield (kg/ha)	1,440.0	1,666.2	+226.2	+15.71%
Harvested Area (ha)	1,500,000	2,414,536	+914,536	+60.97%

Key Insight: Expansion vs. Productivity

While total output grew by 86.25%, the majority of this growth was enabled by bringing 914,536 additional hectares under cultivation (+60.97%). Yield gains per unit area grew at a much more modest pace (+15.71%).

2.3 Production Change Decomposition

To isolate the structural components driving production growth, the total change in production (ΔP) was decomposed using the accounting identity $P = A \times Y$:

$$\Delta P = (Y_0 \times \Delta A) + (A_0 \times \Delta Y) + (\Delta A \times \Delta Y)$$

Decomposition Component	Formula Contribution	Value (tonnes)	Percentage Contribution
Area Effect ($\Delta A \times Y_0$)	Land Expansion Impact	1,316,932	70.68%
Yield Effect ($\Delta Y \times A_0$)	Productivity Gain Impact	339,300	18.21%
Interaction Effect ($\Delta A \times \Delta Y$)	Combined Area-Yield Growth	206,868	11.10%
Total Production Shift	ΔP Total	1,863,100	100.00%

The quantitative breakdown proves that approximately **70.68%** of net production growth stemmed directly from horizontal land expansion (extensification), whereas technological and management productivity enhancements (intensification) accounted for only **18.21%**.

3. Climate Associations & Extremes

3.1 Correlation Matrix

Pearson correlation coefficients (r) were computed to evaluate the strength and direction of linear relationships between environmental inputs and agricultural outputs.

Variable	Yield	Production	Area	Temperature	Precipitation
Yield	1.00	0.49	-0.01	-0.22	0.61
Production	0.49	1.00	0.86	0.41	0.58
Area	-0.01	0.86	1.00	0.63	0.30
Temperature	-0.22	0.41	0.63	1.00	-0.15
Precipitation	0.61	0.58	0.30	-0.15	1.00

3.2 Analysis of Environmental Drivers

- **Precipitation vs. Yield ($r = 0.61$):** Demonstrates a clear, moderate positive relationship. Higher seasonal rainfall directly supports higher per-hectare productivity in rainfed systems.
- **Temperature vs. Yield ($r = -0.22$):** Displays a weak negative relationship, aligning with agronomic observations that elevated thermal stress during flowering/grain filling accelerates crop maturity and reduces yield potential.

- **Area vs. Production ($r = 0.86$):** Strong linear correlation confirming land expansion as the primary operational driver of national volume.

3.3 Production Extremes & Outlier Assessment

An interquartile range (IQR) assessment confirmed zero statistical outliers, indicating that all historical annual figures fall within normal distribution bounds.

- **Record Production Year (2023):** Reached an all-time high of **4,285,206 tonnes** (yield: **1,763.4~kg/ha**; temperature: **25.66°C**; precipitation: **707.85~mm**).
- **Lowest Production Year (2000):** Recorded a low of **2,160,000 tonnes** (yield: **1,440.0~kg/ha**; temperature: **25.16°C**; precipitation: **477.52~mm**).

4. Discussion & Strategic Implications

The findings indicate that Kenya's maize growth model over the past two decades has relied heavily on land extensification rather than sustainable yield intensification. While total production has nearly doubled, relying on area expansion presents severe future challenges due to land fragmentation, competing land-use demands, and ecological constraints in high-potential agricultural zones.

Furthermore, the strong positive reliance of yields on annual rainfall ($r = 0.61$) highlights ongoing vulnerability to climate variability and drought cycles. As climate change increases precipitation volatility, relying primarily on rainfed land expansion without improving per-hectare productivity increases systemic food security risks.

5. Key Recommendations

1. **Transition to Sustainable Intensification:** Reallocate agricultural policy focus toward increasing per-hectare yields through high-yielding climate-resilient seed varieties, balanced fertilizer application, and modern agronomic extension services.
2. **Water Management & Irrigation Infrastructure:** Expand smallholder supplementary irrigation and rainwater harvesting systems to decouple yield performance from unpredictable seasonal precipitation patterns.
3. **Advanced Analytics & Early Warning:** Expand interactive analytical tools (e.g., the Power BI dashboard developed for this project) to integrate county-level microclimate data, satellite vegetation indices (NDVI), and predictive crop yield modeling.

Data References & Sources

- Food and Agriculture Organization. (2023). *FAOSTAT statistical database: Crop production and harvested area data for Kenya (2000–2024)*. FAO, Rome. <http://www.fao.org/faostat/>
- World Bank Group. (2024). *Climate Change Knowledge Portal (CCKP): Historical climate baseline data for Kenya (2000–2024)*. World Bank, Washington, DC. <https://climateknowledgeportal.worldbank.org/>
- Intergovernmental Panel on Climate Change (IPCC). (2014). *Climate change 2014: Impacts, adaptation, and vulnerability*. Cambridge University Press.
- Kenya National Bureau of Statistics (KNBS). (2022). *Economic survey and agriculture sector statistical report*. Government of Kenya, Nairobi.

Lobell, D. B., & Field, C. B. (2007). Global scale climate-crop yield relationships and the impacts of recent warming. *Environmental Research Letters*, 2(1), 014002.